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Computer Information Technology BS

[College of Sciences](#) / [Computer Science and Cybersecurity](#)
[Undergraduate major](#) / Bachelor of Science

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Courses and Requirements

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Requirements for completing the Computer Information Technology major

- All newly admitted students will be assigned academic advisors and are placed in pre-major status until they have met the program eligibility requirements. The Undergraduate Program Declaration Form must be submitted for a student to be considered for acceptance into the major.
- All courses in the major must be completed with a grade of C- or better.
- At least 16 credits of major requirements must be completed at Metro State University.
- At least 28 credits of upper-division (i.e., 300-level, or higher) coursework must be completed in the major.
- Read and understand the guidelines for transfer courses and prerequisites as listed in the General Guidelines section.

MAJOR REQUIREMENTS

+ [Premajor Foundation \(20 credits\)](#)

To declare the Computer Information Technology major, students are required to complete the following premajor foundation courses with a grade of C- or higher and a minimum GPA of 2.5 for ICS 141 and MATH 215 or transfer equivalents. For further details, reference the General Guidelines section below.

Choose one of the following two courses.

- MATH 115 College Algebra

4 credits

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- MATH 120 Precalculus

4 credits

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Choose one of the following two courses.

- ICS 140 Computational Thinking with Programming

4 credits

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- ICS 265 C Programming

4 credits

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Complete all of the following three courses.

- MATH 215 Discrete Mathematics

4 credits

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- ICS 141 Problem Solving with Programming

4 credits

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- ICS 225 Web Design and Implementation

4 credits

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+ Core (40 credits)

Once the premajor foundation are complete, students must complete all of the following courses with a grade of C- or higher.

The Capstone course, ICS 499, should be taken in the final semester of your program, or at least during the semester you complete the last of the other required major courses.

Students who haven not declared their major or have not been accepted into the major will not be allowed to take any 400-level major courses. For further details on prerequisites, reference the General Guidelines section below.

- CFS 262 Computer and Operating Systems Fundamentals I

4 credits

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- CFS 264 Computer and Operating Systems Fundamentals II

4 credits

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- CYBR 332 Computer Security

4 credits

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- CYBR 362 Networking Protocols and Analysis

4 credits

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- ICS 240 Introduction to Data Structures

4 credits

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- ICS 311 Database Management Systems

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4 credits

- ICS 370 Software Design Models
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4 credits

- ICS 377 Interaction Design for User Experience
- ^

4 credits

- ICS 499 Software Engineering and Capstone Project
- ^

4 credits

Choose one of the following two courses. The other course may be taken as a major elective.

- ICS 325 Internet Application Development
- ^

4 credits

- ICS 425 Model View Controller Architecture-based Web Application
- ^

4 credits

+ Electives (8 credits) or Approved Minor

Either eight upper-division credits, or a minor in a field approved by the academic advisor, are required for the Computer Information Technology major.

Electives Option 8 credits

Complete a minimum of eight upper-division credits (i.e, 300-level, or higher) of elective courses. Consult with academic advisor on acceptable electives.

Note the following:

- At least 4 credits of electives must come from upper-division (i.e., 300-level, or higher) ICS courses, not already required for the major, with the following exceptions: ICS 350I, ICS 372, ICS 460, ICS 38*, and ICS 48*.
- The contents of ICS 490 Special Topics in Information and Computer Sciences and ICS 492 Seminar on Emerging Technologies vary from semester to semester and may be taken more than once for elective credits (with permission of the CSC department) as long as they cover different topics.
- Any 300-level or higher CFS, CYBR, MATH, or STAT course may be used as an elective except for the following courses: CFS 350I, CFS 499, CYBR 350I, CYBR 498, CYBR 499, ICS 372, ICS 460, MATH 350I, MATH 499 and STAT 350I.
- A maximum of 4 credits in ICS 350I Individualized Internship spread over 1-3 semesters may be used as elective credits.

Minor Option

Work with your academic advisor to assess if the minor option is appropriate for your degree and career plan. Students with a previous bachelor's degree should talk to their advisor about whether that degree can be used in lieu of a minor. Recommended fields for a minor include (but are not limited to): Industrial and Applied Mathematics, Applied Statistics, Computer Forensics, Cybersecurity, Design of User Experience, Entrepreneurship and Innovation, Management Information Systems, Project Management, and Technical Communication.

+ General Guidelines

Transfer Courses

Transfer coursework equivalency is determined by the Computer Science and Cybersecurity (CSC) department and is initially evaluated upon admission with updates documented on the Degree Audit Report (DARS). When transferring coursework, please be aware that many universities, community, and technical colleges offer courses equivalent to some of our pre-major courses. Sometimes a course at the lower division at another university or college is equivalent to one of our upper-division courses, or an upper-division course at another institution is equivalent to one of our lower-division courses. To calculate upper-division credits for the major electives or for university graduation requirements, the status of the course at the institution where the student took the course is what matters.

Prerequisites

Students must be aware of and abide by prerequisites for all courses for which they are enrolled. No student may be enrolled in a course unless they have completed all course prerequisites with a grade of C- or higher. Students will be administratively dropped from a course if they have not met the required prerequisites. For some courses, prerequisites are enforced automatically by the registration system. If your DARS report shows you have met the prerequisites for a course, and the registration system will not let you register, please contact your academic advisor.

Coming up...

Undergraduate Open House

Thursday, May 29, 2025
4:00 pm - 6:00 pm

Current Students

Find your advisor in the Computer Information Technology BS program

SCIENCES ADVISING

 [Program planner](#)

Faculty

Connect with faculty in the department of Computer Science and Cybersecurity

FACULTY

Connect with us

Metro State on social media

